

using historical market data, whereas actuarial methods will be more willing to incorporate updated market return expectations as the spending plan is updated regularly throughout retirement. With this overview, we now consider ten strategies in greater detail.

Bengen's Constant Inflation-Adjusted Spending Strategy

The first method to be tested is the original constant inflation-adjusted withdrawal strategy introduced in William Bengen's 1994 article, "Determining Withdrawal Rates Using Historical Data." This will serve as a baseline for subsequent comparison with other strategies. Bengen's rule says to adjust spending annually for inflation and maintain constant inflation-adjusted spending until the portfolio depletes. Annual spending increases by the previous year's inflation rate and does not otherwise adjust for market returns or the present size of the underlying investment portfolio. The withdrawal rate is defined only in terms of the first year of the plan, and subsequent spending rates are no longer tracked as an input to guide spending. Spending continues to adjust for inflation, regardless of portfolio size, unless and until the underlying portfolio is fully depleted.

We have already analyzed how this method performs in historical data. A 4 percent initial spending rate survived all of the rolling historical periods when using a 50/50 portfolio. Exhibit 6.5 shows how this plays out, with sixty-two lines (on top of one another, in this case) with real spending of \$4 from an initial portfolio of \$100, reflecting the sixty-two rolling historical thirty-year periods in the data set for which the rule could provide a consistent real income.

The bottom part of the exhibit shows remaining wealth throughout retirement for each of the sixty-two rolling historical periods in the data set. In contrast with the smooth income provided, remaining wealth is much more volatile. It runs as low as \$3.36 after thirty years for the 1966 retiree (whose portfolio will deplete in year thirty-one) and up to \$463.9 after thirty years for the 1982 retiree. The volatility in remaining wealth is a byproduct of keeping spending at the same level regardless of portfolio performance. (These numbers are all scalable, as you can just multiply the assumed \$100 initial wealth by the factor needed to obtain actual wealth to know the implications for spending and remaining wealth from a different starting